



DEPI GRADUATION PROJECT

Frontend Web Development

Lexi

Learn Languages Naturally

A Neuro-Based Language Learning Platform

The Lexi logo, featuring the word "Lexi" in a dark blue, handwritten-style font, followed by a stylized bee icon. The bee is also in dark blue and has a segmented body, wings, and antennae. The logo is centered within a white circle that has a thin blue border. The circle is set against a dark blue background with faint yellow curved lines.

THE TEAM

Meet the People Behind Lexi

A student team from DEPI, united by one goal — making language learning feel natural.



Nour Ayman

Team Member



Aisha Mahmoud

Team Member



Marwa Ahmed

Team Member



Shahd Medhat

Team Member

THE PROBLEM

Why is learning a new language so difficult?

Millions of learners struggle to retain vocabulary because traditional learning methods often rely on memorization rather than real understanding.



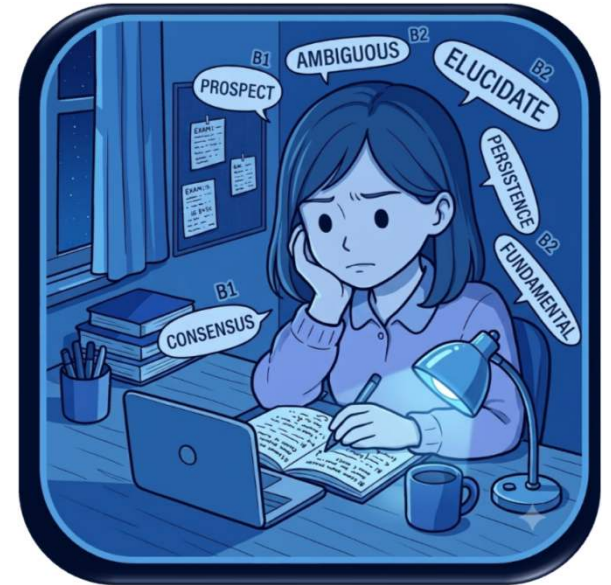
Forgetting newly learned words quickly



Learning vocabulary without meaningful context



Losing motivation after repetitive practice



THE LANDSCAPE

Current Learning Methods

Most language learning platforms provide valuable resources, but many still focus on isolated vocabulary and repetitive exercises.



Words are learned outside their real-life context



Practice is not personalized



Learning becomes repetitive



OUR VISION

Our Vision

We believe language learning should happen naturally through daily interaction rather than memorization.

Lexi transforms everyday reading and watching into learning opportunities by helping users:

- ✓ Discover
- ✓ Understand
- ✓ Remember vocabulary in context.



INTRODUCING

Meet Lexi

Lexi is a neuro-based language learning platform that helps users acquire vocabulary naturally. Instead of memorizing lists, Lexi encourages learning through:

Context

Daily Exposure

Adaptive Practice

Personalized Recommendations

"Learning becomes a by-product of using the language."



THE JOURNEY

How Lexi Works



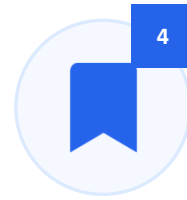
Read / Watch



Find Unknown Word



Look It Up



Save It



Practice Over Time



Master It Naturally

Core Learning Experience

How Lexi transforms language learning into a natural daily habit.

CORE FEATURE

Smart Lookup



Quickly look up any word or phrase while reading.

- Simple translations
- Multiple real-life contexts
- Example sentences
- Save words instantly

Adaptive Mastery



Lexi adapts practice to each learner instead of repeating the same exercises.



CORE FEATURE

Vault



Every saved word is organized in one place.

- Browse saved vocabulary
- Review previous words
- Read personalized paragraphs

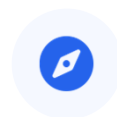
Watch & Learn



Learn naturally while watching videos.

- Interactive captions
- Instant word lookup
- Save vocabulary directly from videos

Explore



Discover new vocabulary based on your learning journey.

- Previously saved vocabulary
- Personal interests
- Learning history

COMPARISON

Why Choose Lexi?

Traditional Learning

- ✗ Memorize vocabulary
- ✗ Fixed learning path
- ✗ Passive repetition
- ✗ Separate learning from daily life
- ✗ Easy to forget

Lexi

- ✓ Learn through context
- ✓ Personalized experience
- ✓ Adaptive practice
- ✓ Daily learning
- ✓ Long-term retention

Lexi transforms language learning from a chore into a natural daily habit.

Business Feasibility & Growth Strategy

How can Lexi evolve from a graduation project into a sustainable language learning platform?

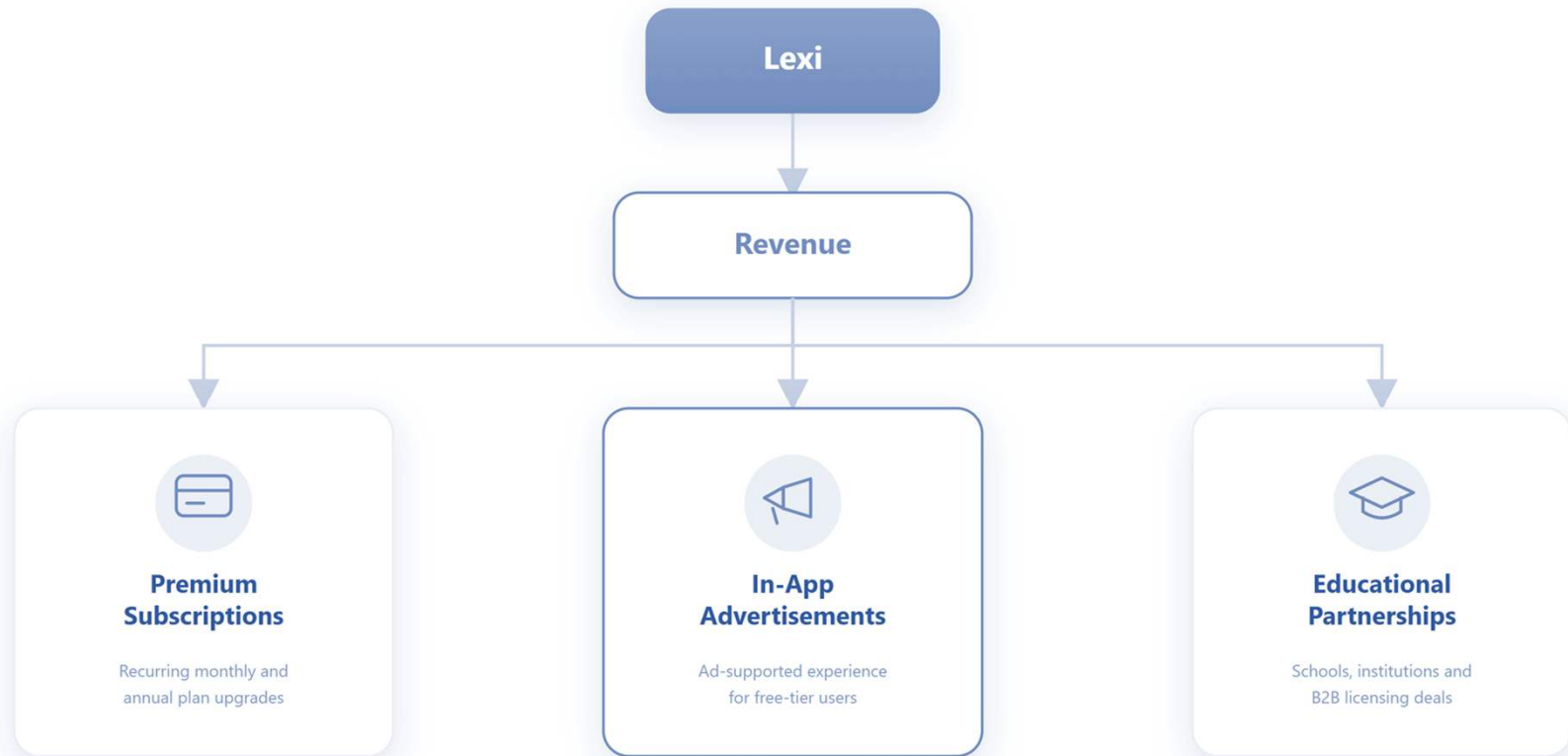
BUSINESS MODEL

Revenue Model (Value-Driven Pricing)

Free	Lexi Pro	Lexi Ultra
Get started with the basics	For committed learners	AI-powered mastery
\$0 per month	\$7 per month or \$60/year (\$5/mo equivalent)	\$14 per month or \$120/year (\$10/mo equivalent)
✓ 30 contextual word lookups ✓ Standard adaptive learning ✓ Limited video learning ✓ Basic Vault access ✓ Ad-supported ✓ Community support	✓ Unlimited lookups ✓ Unlimited video learning ✓ Full Vault ✓ Personalized learning progress ✓ Advanced analytics ✓ Ad-free experience	✓ AI-generated reading content ✓ AI context explanations ✓ AI recommendations ✓ Intelligent study planning ✓ Everything in Lexi Pro
		
Get Started Free	Start Lexi Pro	Go Ultra

BUSINESS MODEL

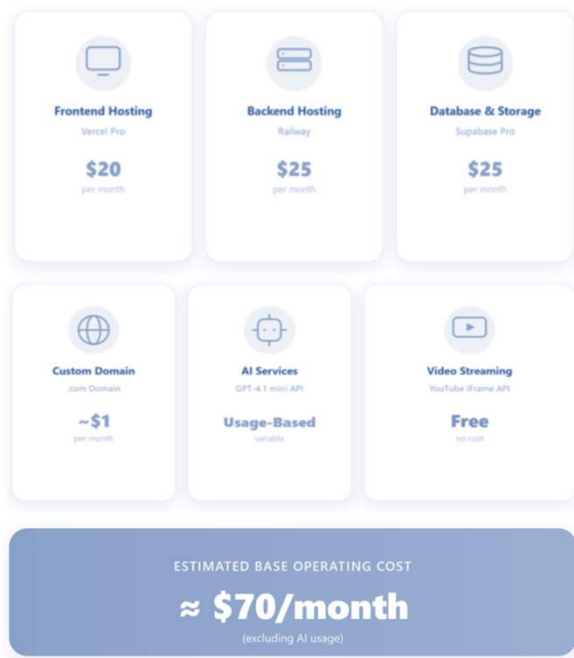
Revenue Streams



FINANCIAL MODEL

Operating Cost & Revenue Model

Cost-Efficient Infrastructure



Revenue Sources

Revenue Sources



Free Users

Advertisement Revenue · Premium Conversion



Premium Users

Monthly Subscription · Annual Subscription

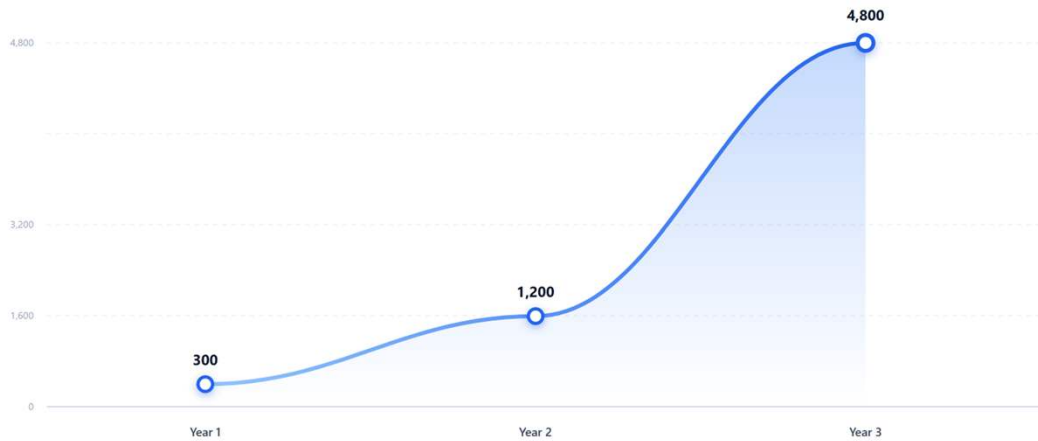
Break-even Estimate

~25–30 Premium Subscribers

Estimated based on projected infrastructure costs and expected subscription adoption.

FINANCIAL PROJECTIONS

Projected Growth Based on User Adoption



Assumptions

- Premium subscription: \$7/month
- Estimated conversion rate: 3–4%
- Growth driven by increasing user adoption

Year 1

10,000 Active Users
300 Premium Subscribers

Year 2

40,000 Active Users
1,200 Premium Subscribers

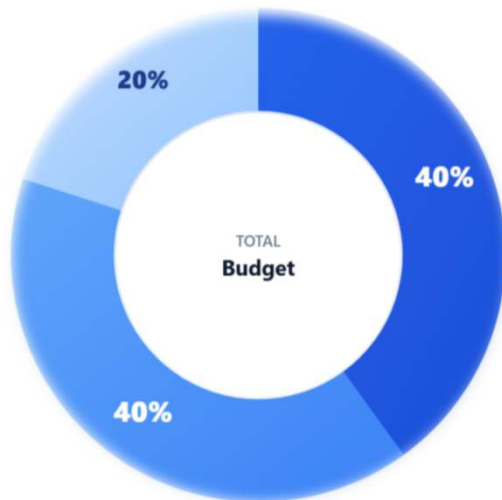
Year 3

120,000 Active Users
4,800 Premium Subscribers

Projected growth assumes continuous product improvements, marketing efforts, and increasing user engagement.

INVESTMENT PLAN

Strategic Resource Allocation



Product Development (40%)

- Platform optimization
- Backend improvements
- API integrations
- Performance enhancements
- Future AI integration

Marketing & User Growth (40%)

- Digital marketing campaigns
- University outreach
- Community building
- User acquisition initiatives

Operations & Infrastructure (20%)

- Cloud infrastructure
- Security
- Platform maintenance
- Legal & compliance preparation

This allocation is designed to support Lexi's projected growth strategy while maintaining a balanced investment across product development, user acquisition, and operational sustainability.

Quality Assurance, Testing & Reliability

Ensuring Lexi delivers a stable, secure, and production-ready learning experience.

LAUNCH READINESS

Executive Summary & Launch Readiness

Lexi is Ready for Deployment



Zero Critical Defects

- No critical issues were identified during functional, integration, and edge-case testing.
- 100% pass rate across all 16 critical testing modules — no critical issues found in functional, integration, or edge-case testing.



Deployment Ready

- The platform is stable, responsive, and ready for real-world deployment.
- Verified stable functionality, secure API integration, and consistent behavior across platforms and devices.



Reliable User Experience

- Core learning features perform consistently under normal and stress-test conditions.
- Core user journeys and AI-powered features run flawlessly, even under simulated system stress

Testing Metrics Dashboard

Metric	Value	Status
Total test scenarios	16	Complete
Passed cases	16	Successful
Failed cases	0	No defects found
Blocked cases	0	No blockers
Overall pass rate	100%	Target achieved



100% Test Pass Rate

Comprehensive Test Results Matrix

Module	Test scenario / injection	Observed result and state management	Status
Vault	Low seed data (limited saved vocabulary)	AI heuristics accurately interpolate and generate contextual content with minimal input tokens.	✓ Pass
Time management	Extreme timezone / UTC changes	Chronological tracking layers preserve original state data accurately regardless of client offset.	✓ Pass
Watch module	Unsupported video/language parsing	System validates video metadata dynamically and filters out unsupported assets before rendering.	✓ Pass
Video captions	Missing/malformed caption files	Script parsing flags the error and skips the invalid asset automatically; player queue unblocks.	✓ Pass
Recommendation	Cold-start state (zero historical data)	Default generalized recommendation matrices load flawlessly on initial profile execution.	✓ Pass
Explore module	Miniature or zero-record dataset mapping	Render logic maps zero-length arrays safely without throwing null pointer or indexing errors.	✓ Pass
Search validation	Ingestion of harmful or unsafe queries	Input field sanitization layer catches, blocks, and safely flags malicious strings before database write.	✓ Pass
AI question gen.	LLM service generation failure	System intercepts timeout and falls back deterministically to static, cached backup MCQs.	✓ Pass

RELIABILITY

Architectural & Core Engineering Findings

Designed for Stable and Reliable Performance



Network Recovery

Network Resilience.
Handles unstable connections.
Uses cached content when needed.



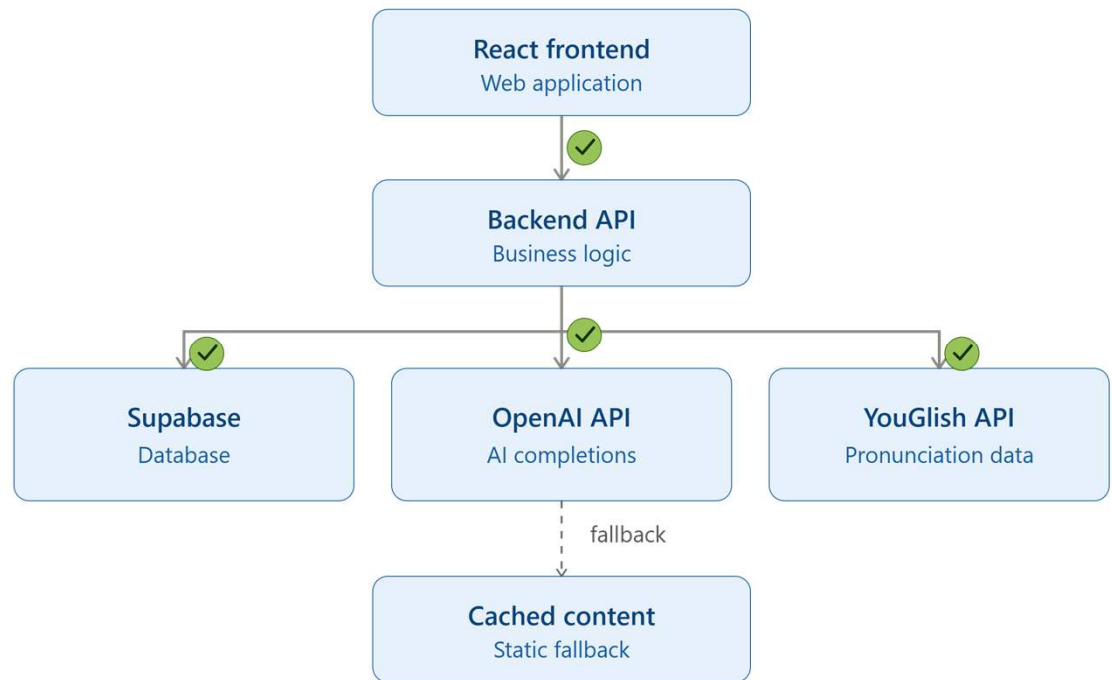
External API Stability

Dependency Isolation.
Backend isolates external APIs.
Uses fallback if an API fails.



Data Consistency

Input Security
Validates and sanitizes user input.
Protects the backend from invalid requests.



FAULT TOLERANCE

AI Dependency & Fallback Testing

Reliable Recovery from External Service Failures

Recovery Execution	Failure Injection / Stress Test	Integration Focus
Switched to backup MCQs	AI Timeout	LLM Generation
Gracefully handled empty results	Null response / Empty payload	YouGlish API
Skipped invalid media safely	Missing or malformed captions	Video Processing
Cold-start executed successfully	Empty user history	Recommendation Engine

DATA INTEGRITY

Validation & Edge Case Testing

Ensuring Accurate Data Handling



Input Validation

User inputs are validated to prevent unexpected behavior.



Data Sanitization

Invalid characters and malformed queries are safely filtered.



Boundary Testing

System correctly handles limits and edge values.



Performance Under Load

Large learning datasets remain responsive without freezing.

QUALITY MODEL

Software Quality Factors

Delivering a High-Quality Learning Experience



Performance

Fast loading and smooth interaction.



Security

Protected user data and secure authentication.



Usability

Simple and intuitive learning experience.



Compatibility

Consistent performance across devices and browsers.



Reliability

Stable behavior under different usage conditions.



Maintainability

Modular architecture for easier future development.

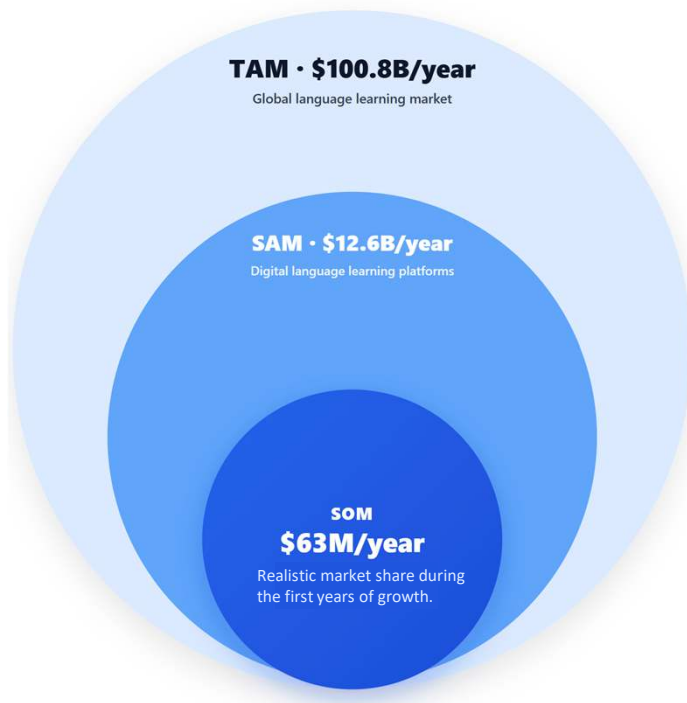
Market Analysis & GTM

Identifying the right audience and delivering Lexi where it matters most.

MARKET

Market Opportunity

A large and growing market with significant potential for AI-powered language learning.



TAM – Total Addressable Market

\$100.8B/yr

1.2 billion users learning globally, at \$84/year average value.

SAM – Serviceable Available Market

\$12.6B/yr

150 million users actively learning via smart mobile apps.

SOM – Serviceable Obtainable Market

\$63M/yr

A realistic 0.5% SAM capture within 2 – 3 years – 750K paid subscribers.

COMPETITIVE LANDSCAPE

Competitive Analysis

How Lexi differentiates itself from existing language learning platforms.

	Lexi	Duolingo	Babbel	Rosetta Stone	Memrise	HelloTalk
Context-based Learning	✓	—	~	—	—	~
Native Video Learning	✓	—	—	—	~	—
AI Personalization	✓	~	—	—	~	—
Vocabulary Vault	✓	—	—	—	—	—

Contextual Intelligence

The gap left by Duolingo & Memrise.

Native Video Engagement

Where Babbel & Rosetta Stone fall short.

AI Personalization

Entirely unique to Lexi.

While competitors focus on gamified streaks or strict grammar rules, Lexi focuses on natural acquisition – making language learning a byproduct of everyday digital life – not a chore.



CUSTOMERS

Target Audience & Customer Segments

Designed for learners who want to improve English naturally through real-world experiences.



University Students

Improve academic communication and everyday English.



Travelers

Learn practical English for real-life situations.



Movie & Content Enthusiasts

Understand authentic English through videos and media.



Young Professionals

Strengthen workplace communication and career opportunities.

Why They Choose Lexi



Learn through real-world content



AI-powered personalized learning



Context instead of memorization

ROADMAP

Go-To-Market Strategy

A phased roadmap for sustainable product growth.



PHASE 1

Soft Launch & Early Adopters

- Closed beta – 1,000 users
- University students, travelers, Netflix-culture learners
- Stress-test the Vault & Watch core loop
- Structured feedback loop feeds products iteration



PHASE 2

Growth & Virality

- TikTok & Instagram content: on words used differently in movies vs. real life
- Micro-influencer partnerships (20K – 50K followers) in language education
- PLG: Share custom “Vault Read” essays to Instagram stories with a Lexi watermark
- Community building & engagement



PHASE 3

Monetization & Retention

- Premium subscriptions (Pro & Ultra)
- Personalized notifications
- Freemium funnel: free caps on Mastery & Watch
- Long-term customer retention: predictive push notifications timed to the forgetting curve



Thank You!

Thank you for your time and attention.
We appreciate the opportunity to present Lexi.

Questions?